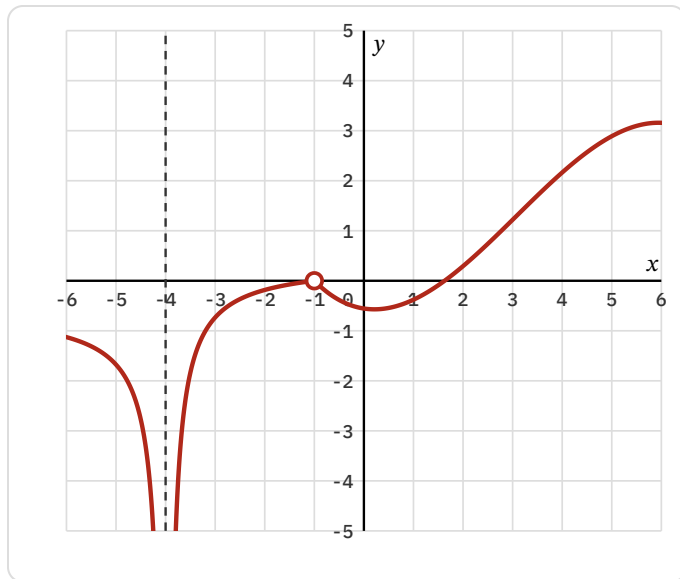


Mock Midterm A — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 92 points

1. Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE. 8 pts



- a.) $f(-4)$ (1)
DNE
- b.) $\lim_{x \rightarrow -4^-} f(x)$ (1)
 $-\infty$
- c.) $\lim_{x \rightarrow -4^+} f(x)$ (1)
 $-\infty$
- d.) $\lim_{x \rightarrow -4} f(x)$ (1)
 $-\infty$
- e.) $f(-1)$ (1)
DNE
- f.) $\lim_{x \rightarrow -1^-} f(x)$ (1)
0
- g.) $\lim_{x \rightarrow -1^+} f(x)$ (1)
0
- h.) $\lim_{x \rightarrow -1} f(x)$ (1)
0

WORKED SOLUTION

At $x = -4$

There is no filled dot on the line $x = -4$, so $f(-4)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = -4$ downward, so the limit is $-\infty$.

From the right, the curve runs along the asymptote $x = -4$ downward, so the limit is $-\infty$.

Both sides go to $-\infty$, so $\lim_{x \rightarrow -4} f(x) = -\infty$ (unbounded, so no finite limit exists).

At $x = -1$

There is no filled dot on the line $x = -1$, so $f(-1)$ is undefined $\rightarrow$ DNE.

Tracing the curve toward $x = -1$ from the left, the heights approach 0 (open or closed dot doesn't matter).

Tracing the curve toward $x = -1$ from the right, the heights approach 0 (open or closed dot doesn't matter).

Both one-sided limits equal 0, so $\lim_{x \rightarrow -1} f(x) = 0$ — even though $f(-1)$ is undefined. The limit ignores the value AT the point.

2. Let $f(x) = \frac{\sqrt{x}-4}{x-16}$. We want to know what happens to the outputs when x is near **16**.

7 pts

x	15.9	15.99	15.999	16	16.001	16.01	16.1
$f(x)$	0.125196	0.12502	0.125002	?	0.124998	0.12498	0.124805

a.) Can you find $\lim_{x \rightarrow 16} f(x)$ by plugging in $x = 16$? (1)

(A) No — substituting gives $\frac{0}{0}$, which is undefined

b.) As $x \rightarrow 16^-$, $f(x)$ approaches... (2)

$$\frac{1}{8}$$

c.) As $x \rightarrow 16^+$, $f(x)$ approaches... (2)

$$\frac{1}{8}$$

d.) $\lim_{x \rightarrow 16} f(x) =$ (2)

$$\frac{1}{8}$$

WORKED SOLUTION

Plugging in $x = 16$ makes both the top and bottom 0, so we can't substitute.

Reading the table left to right, the outputs close in on $\frac{1}{8}$ from both sides (the values at 16 ± 0.001 are the closest).

$$\lim_{x \rightarrow 16} \frac{\sqrt{x}-4}{x-16} = \frac{1}{8}$$

Algebra check: $\frac{\sqrt{x}-4}{(\sqrt{x}-4)(\sqrt{x}+4)} = \frac{1}{\sqrt{x}+4} \rightarrow \frac{1}{8}$

3. Suppose

4 pts

$$\lim_{x \rightarrow 4} f(x) = 1, \quad \lim_{x \rightarrow 4} g(x) = -4, \quad \lim_{x \rightarrow 4} h(x) = 25$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 4} ([f(x)]^3 - g(x))$ (1)

5

b.) $\lim_{x \rightarrow 4} \sqrt{h(x)} f(x)$ (1)

5

c.) $\lim_{x \rightarrow 4} \frac{g(x)-2}{f(x)}$ (1)

-6

d.) $\lim_{x \rightarrow 4} x^2 f(x)$ (1)

16

WORKED SOLUTION

a. $\lim_{x \rightarrow 4} ([f(x)]^3 - g(x)) = (1)^3 - (-4) = 1 + 4 = 5$

b. $\lim_{x \rightarrow 4} \sqrt{h(x)} f(x) = \sqrt{25} \cdot (1) = 5 \cdot (1) = 5$

c. $\lim_{x \rightarrow 4} \frac{g(x)-2}{f(x)} = \frac{-4-2}{1} = -6$

d. $\lim_{x \rightarrow 4} x^2 f(x) = (4)^2 \cdot (1) = 16$

The quotient property works here because the bottom limit isn't 0.

4. Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE. 4 pts

a.) $\lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{x-25}$ (2)

$\frac{1}{10}$

b.) $\lim_{h \rightarrow 0} \frac{\frac{1}{4+h} - \frac{1}{4}}{h}$ (2)

$-\frac{1}{16}$

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{x-25} = \lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{(\sqrt{x}-5)(\sqrt{x}+5)} = \lim_{x \rightarrow 25} \frac{1}{\sqrt{x}+5} = \frac{1}{5+5} = \frac{1}{10}$$

b.

$$\lim_{h \rightarrow 0} \frac{\frac{1}{4+h} - \frac{1}{4}}{h} = \lim_{h \rightarrow 0} \frac{\frac{4-(4+h)}{4(4+h)}}{h} = \lim_{h \rightarrow 0} \frac{-h}{4(4+h)h} = \lim_{h \rightarrow 0} \frac{-1}{4(4+h)} = -\frac{1}{16}$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use $\infty / -\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow \infty} \frac{-3x^3 - x^2}{5x}$ (2)
 $-\infty$

b.) $\lim_{x \rightarrow \infty} \frac{7x+2}{-5x-5}$ (2)
 $-\frac{7}{5}$

c.) $\lim_{x \rightarrow \infty} \frac{3x^3}{-5x^4 - 5x^3 + 7x^2}$ (2)
 0

WORKED SOLUTION

a. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-3x^3 - x^2}{5x} = \lim_{x \rightarrow \infty} \frac{-3x^2 - x}{5}$$

The bottom $\rightarrow 5$, but the top is dominated by $-3x^2$, which $\rightarrow -\infty$ as $x \rightarrow \infty$; dividing by 5 gives $-\infty$.

b. Divide every term by x , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{7x + 2}{-5x - 5} = \lim_{x \rightarrow \infty} \frac{7 + \frac{2}{x}}{-5 - \frac{5}{x}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{7}{-5} = -\frac{7}{5}$.

c. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{3x^3}{-5x^4 - 5x^3 + 7x^2} = \lim_{x \rightarrow \infty} \frac{\frac{3}{x}}{-5 - \frac{5}{x} + \frac{7}{x^2}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -5$, so the limit is 0 .

6. Let

7 pts

$$G(x) = \begin{cases} x^2 - 2 & x < -2 \\ -x & x > -2 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(-2)$ (1)

DNE

b.) $\lim_{x \rightarrow -2^-} G(x)$ (1)

2

c.) $\lim_{x \rightarrow -2^+} G(x)$ (1)

2

d.) $\lim_{x \rightarrow -2} G(x)$ (1)

2

e.) $G(1)$ (1)

-1

f.) $\lim_{x \rightarrow -\infty} G(x)$ (1)

∞

g.) $\lim_{x \rightarrow \infty} G(x)$ (1)

$-\infty$

WORKED SOLUTION

$G(-2)$: the pieces cover $x < -2$ and $x > -2$ only, so $G(-2)$ is undefined $\rightarrow$ DNE.

Left: use $x^2 - 2$: $(-2)^2 - 2 = 2$.

Right: use $-x$: $-1(-2) + 0 = 2$.

Two-sided: both sides give 2, so the limit is 2 (even though $G(-2)$ is undefined).

$G(1)$: $1 > -2$, so use $-x$: $-1(1) + 0 = -1$.

End behavior: as $x \rightarrow -\infty$ we're on the $x^2 - 2$ piece — an even-degree term with positive leading coefficient $\rightarrow \infty$. As $x \rightarrow \infty$ we're on the line with slope -1 $\rightarrow -\infty$.

7. Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 - 5x + 3}{2x - 3} & x < \frac{3}{2} \\ 4x + 4 & x \geq \frac{3}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 5} f(x)$ (1)

24

b.) $\lim_{x \rightarrow -2} f(x)$ (1)

-3

c.) $f\left(\frac{3}{2}\right)$ (1)

10

d.) $\lim_{x \rightarrow \frac{3}{2}^-} f(x)$ (1)

$\frac{1}{2}$

e.) $\lim_{x \rightarrow \frac{3}{2}^+} f(x)$ (1)

10

f.) $\lim_{x \rightarrow \frac{3}{2}} f(x)$ (1)

DNE

WORKED SOLUTION

For $x < \frac{3}{2}$, factor the top: $\frac{(2x-3)(x-1)}{2x-3} = x - 1$ (valid since $x \neq \frac{3}{2}$ there).

a. $5 > \frac{3}{2} \rightarrow$ bottom piece: $4(5) + 4 = 24$.

b. $-2 < \frac{3}{2} \rightarrow$ top piece, simplified: $-2 - 1 = -3$.

c. The bottom piece includes $x = \frac{3}{2}$ ($\geq$), so $f\left(\frac{3}{2}\right) = 4\left(\frac{3}{2}\right) + 4 = 10$.

d. Left: $\lim_{x \rightarrow \frac{3}{2}^-} (x - 1) = \frac{1}{2}$.

e. Right: $\lim_{x \rightarrow \frac{3}{2}^+} (4x + 4) = 10$.

f. $\frac{1}{2} \neq 10$, so the limit DNE.

8. Is $f(x)$ continuous at $x = -1$?

5 pts

$$f(x) = \begin{cases} \frac{x^2 + 6x + 5}{x + 1} & x \neq -1 \\ 4 & x = -1 \end{cases}$$

a.) $f(-1)$ (2)

4

b.) $\lim_{x \rightarrow -1} f(x)$ (2)

4

c.) Is f continuous at $x = -1$? (1)

(A) Yes — $f(-1)$ exists, the limit exists, and they are equal

WORKED SOLUTION

1. $f(-1) = 4$ (given directly by the second piece).

2. $\lim_{x \rightarrow -1} \frac{(x+1)(x+5)}{(x+1)} = \lim_{x \rightarrow -1} (x+5) = 4.$

3. $4 = 4$. All three conditions hold $\rightarrow$ continuous.

9. Classify each function at the given x -value.

3 pts

a.) $f(x) = x^2 + 5x - 4$ at $x = -1$ (1)

(A) Continuous

b.) $f(x) = \frac{x^2-4}{x+2}$ at $x = -2$ (1)

(B) Removable discontinuity

c.) $f(x) = \frac{x+2}{x^2+x-2}$ at $x = -2$ (1)

(B) Removable discontinuity

WORKED SOLUTION

a. Polynomials are continuous everywhere.

b. $0/0$ at -2 , but it cancels to $x - 2$, so the limit exists (-4) while $f(-2)$ is undefined $\rightarrow$ removable.

c. Factor: $\frac{(x+2)}{(x+2)(x-1)}$. The $(x+2)$ cancels, leaving $\frac{1}{x-1} \rightarrow -\frac{1}{3} \rightarrow$ removable.

10. Let $C(q) = q^2 + 3q + 200$, where $C(q)$ represents the cost, in dollars, of producing q items. 7 pts
Find the average rate of change over $[4, 8]$.

a.) $C(4)$ (2)

228

b.) $C(8)$ (2)

288

c.) Average rate of change (number only) (2)

15

d.) Which sentence correctly states the result **with units**? (1)

(B) Between $q = 4$ and $q = 8$ items, the cost changes at an average rate of 15 dollars per item.

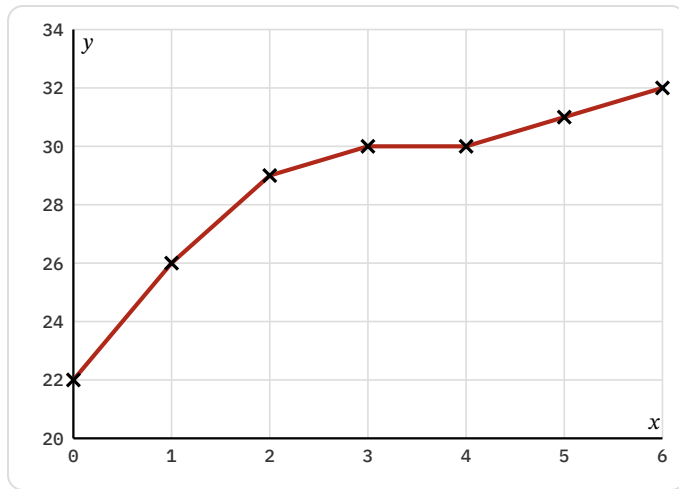
WORKED SOLUTION

$$C(4) = 228, \quad C(8) = 288$$

$$\frac{C(8) - C(4)}{8 - 4} = \frac{288 - (228)}{4} = \frac{60}{4} = 15 \text{ dollars per item}$$

It's an *average* over an interval (not an instant), and the units are dollars per item — dollars per item. The total change would be 60 dollars.

11. The graph shows the closing price (in dollars) of a stock at the start of each month in 2026, where $x = 0$ is January, $x = 1$ is February, and so on. 3 pts



- a.) Average rate of change on $[2, 5]$ (dollars per month) (2)
 $\frac{2}{3}$
- b.) On which interval is the average rate of change greatest? (1)
(A) $[0, 2]$

WORKED SOLUTION

$$\frac{f(5) - f(2)}{5 - 2} = \frac{31 - 29}{3} = \frac{2}{3} \text{ dollars per month}$$

Between Mar and Jun the price changed by $\frac{2}{3}$ dollars per month on average — the slope of the secant line joining those two points.

Greatest-rate check: $[0, 2]: \frac{7}{2}$, $[2, 4]: \frac{1}{2}$, $[4, 6]: 1 \rightarrow [0, 2]$.

12. Let $f(x) = 2x^2 + 2x + 3$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table. 7 pts

a.) Simplified difference quotient (in terms of x and h) (3)

$$4x + 2h + 2$$

b.) $x = 3, h = 1$ (1)

$$16$$

c.) $x = 3, h = 0.1$ (1)

$$\frac{71}{5}$$

d.) $x = 3, h = 0.01$ (1)

$$\frac{701}{50}$$

e.) As $h \rightarrow 0$, what value do these approach? (1)

$$14$$

WORKED SOLUTION

$$f(x+h) = 2(x+h)^2 + 2(x+h) + 3 = 2x^2 + 4xh + 2h^2 + 2x + 2h + 3$$

$$f(x+h) - f(x) = 4xh + 2h^2 + 2h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{h(4x + 2h + 2)}{h} = 4x + 2h + 2$$

Now substitute:

$$x = 3, h = 1 : 16$$

$$x = 3, h = 0.1 : \frac{71}{5} = 14.2$$

$$x = 3, h = 0.01 : \frac{701}{50} = 14.02$$

Letting $h \rightarrow 0$ leaves **14**: the instantaneous rate of change (slope of the tangent line) at $x = 3$. That's exactly what section 1.4 formalizes.

13. Let $f(x) = x^3 - x$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$. 8 pts

a.) $f'(x) =$ (3)
 $3x^2 - 1$

b.) $f'(-2) =$ (2)
 11

c.) Slope of the tangent line to f at $x = 2$ (2)
 11

d.) What does $f'(-2) = 11$ mean? (1)
(A) The instantaneous rate of change of f at $x = -2$ is 11 — the slope of the tangent line there.

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{[(x+h)^3 - 1(x+h)] - (x^3 - x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - 1h}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 1) = 3x^2 - 1 \end{aligned}$$

Then $f'(-2) = 11$ and the slope of the tangent at $x = 2$ is $f'(2) = 11$.

$f'(-2)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. Let $f(x) = -x^2 + x$. Find the equation of the tangent line to the graph of f at $x = -2$. 7 pts

a.) $f(-2)$ (2)
 -6

b.) $f'(-2)$ (2)
 5

c.) Tangent line (3)
 $y = 5x + 4$

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[-(x+h)^2 + (x+h)] - (-x^2 + x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-x^2 - 2xh - h^2 + x + h + x^2 - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2xh - h^2 + h}{h} = \lim_{h \rightarrow 0} \frac{h(-2x - h + 1)}{h} \\ &= \lim_{h \rightarrow 0} (-2x - h + 1) = -2x + 1 \end{aligned}$$

Point: $f(-2) = -6$, so the point is $(-2, -6)$. Slope: $m = f'(-2) = 5$.

$$y + 6 = 5(x + 2) \Rightarrow y = 5x + 4$$

Answers in point-slope form are accepted too.

15. Let $T(t) = -t^2 + 30t + 70$ be the temperature, in °F, of an oven t minutes after it is turned on. 6 pts

a.) $T'(t) =$ (3)
 $-2t + 30$

b.) $T'(12)$ (2)
 6

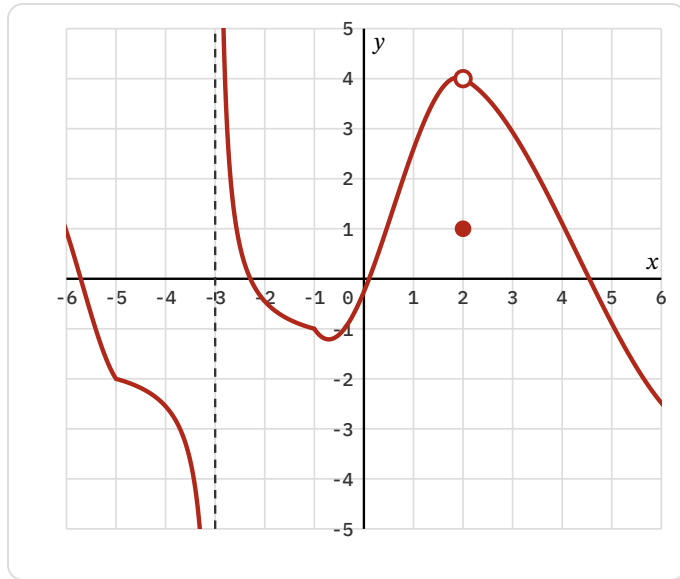
c.) Interpret $T'(12) = 6$. (1)
(A) At $t = 12$ minutes, the oven's temperature is increasing at a rate of 6 degrees Fahrenheit per minute.

WORKED SOLUTION

$$\begin{aligned} T'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-(t+h)^2 + 30(t+h) + 70] - (-t^2 + 30t + 70)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-t^2 - 2th - h^2 + 30t + 30h + 70 + t^2 - 30t - 70}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2th - h^2 + 30h}{h} = \lim_{h \rightarrow 0} \frac{h(-2t - h + 30)}{h} \\ &= \lim_{h \rightarrow 0} (-2t - h + 30) = -2t + 30 \\ T'(12) &= -2(12) + 30 = 6 \end{aligned}$$

Positive/negative tells the direction, and the units are degrees Fahrenheit per minute: at that instant the oven's temperature is increasing at 6 degrees Fahrenheit per minute. (The value $T(12) = 286$ is a different quantity.)

16. At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there. 4 pts



- a.) At $x = -5$ (1)
(B) Continuous but NOT differentiable
- b.) At $x = -3$ (1)
(C) Not continuous (so not differentiable)
- c.) At $x = -1$ (1)
(A) Continuous and differentiable
- d.) At $x = 2$ (1)
(C) Not continuous (so not differentiable)

WORKED SOLUTION

$x = -5$: No break, but the curve comes to a sharp point (corner): the slopes from the left and right disagree $\rightarrow$ continuous, not differentiable.

$x = -3$: Vertical asymptote $\rightarrow$ not continuous, so not differentiable.

$x = -1$: Smooth and unbroken — a single tangent line fits $\rightarrow$ continuous and differentiable.

$x = 2$: There's a hole (the dot isn't on the curve) $\rightarrow$ not continuous, so not differentiable.
